

MUST lecture 1

Multivariate data

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

p-dimensional random variable (random vector)

Vector random variable $X = (X_1, X_2, \dots, X_p)^T$

Population mean vector $\mu = (\mu_1, \dots, \mu_p)^T = (E[X_1], \dots, E[X_p])^T$

Population covariance matrix $\Sigma = E[(X - \mu)(X - \mu)^T]$
 $= \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots \\ \sigma_{12} & \ddots & \vdots \\ \vdots & \dots & \sigma_p^2 \end{bmatrix}$ where

Marginal variances $\sigma_i^2 = V[X_i] = E[(X_i - \mu_i)^2], \quad i = 1, \dots, p$

Pairwise covariances $\sigma_{ij} = \text{Cov}[X_i, X_j] = E[(X_i - \mu_i)(X_j - \mu_j)]$

Population correlation matrix $\rho = \begin{bmatrix} 1 & \rho_{12} & \cdots \\ \rho_{12} & 1 & \vdots \\ \vdots & \dots & 1 \end{bmatrix}, \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_i^2} \sqrt{\sigma_j^2}}$

Sample data

n vector observations, each of dimension (p x 1)

Data matrix (n x p):
$$X = \{X_{ji}\} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \dots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix} \quad j = 1, \dots, n \text{ \& } i = 1, \dots, p$$

Descriptive statistics

Sample mean vector
$$\hat{\mu} = \bar{X} = (\bar{X}_1, \dots, \bar{X}_p)^T \quad E[\hat{\mu}] = \mu \quad Cov[\hat{\mu}] = \frac{1}{n} \Sigma$$

Sample covariance matrix
$$\hat{\Sigma} = S = \frac{1}{n-1} \sum_{j=1}^n (X - \bar{X})(X - \bar{X})^T \quad E[S] = \Sigma$$

Sample covariance matrix

$$\hat{\Sigma} = S = \begin{bmatrix} s_1^2 & s_{12} & \cdots & s_{1p} \\ s_{12} & s_2^2 & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \cdots & s_p^2 \end{bmatrix}$$

Sample correlation matrix

$$\hat{\rho} = R = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{12} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1p} & r_{2p} & \cdots & 1 \end{bmatrix}$$

$$r_{ij} = \frac{s_{ij}}{\sqrt{s_i^2} \sqrt{s_j^2}}$$

Generalized sample variance
Total sample variance

$$|S| = \det(S) = \prod_{i=1}^p \lambda_{i,S}$$
$$\text{trace}(S) = \sum_{i=1}^n s_i^2 = \sum_{i=1}^n \lambda_{i,S}$$

Visualization of (multivariate) data

1D data ($p = 1$): histogram, boxplot

2D data ($p = 2$): marginal histograms and boxplots
scatterplot between variables

pD data (general p): scattermatrix (matrix of plots)
diagonal = marginal histograms or boxplots
off-diagonal = pairwise scatterplots